Team Foundry

CLAY EARTH DATASET FOUNDRY · TRACK 1 · TEAM 46

# An AI agent says it found 15 data centers. Can ADB trust that dataset?

> We built the system that answers that question **with evidence**, step by step and claim by claim.


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OUR IDEA

# Judge the work, not just the answer

### Record

Every model call and every tool call is captured as an OpenTelemetry span: what the agent saw, what it decided, what each tool returned.

### Verify

Every claim must cite the tool result it came from. Code checks the value is really there, and that each site is inside the area that was asked for.

### Judge

Three AI judges from three different companies score the parts code can't, each vote tied to the step it relied on.

Nothing is judged on the final answer alone. The trace is the evidence.

LIVE

# Judging a dataset the agent already built

1. **The run:** "data centers within 10 miles of national or state parks in Texas", already built by Kimi K3 on AgentCore
2. **Judge again:** three judges start. Kimi steps down for its own agent; Mistral sits.
3. **While they think:** 15 / 15 claims verified against the tools, 8 / 8 sites inside Texas, on the map
4. **The verdicts land:** a score and how far to trust it
5. **Every vote points at evidence:** click a judge's reason, land on the tool call

▶ Judge this run live

Contrast: a run that fails

Backup: India solar parks

The score moves a little each time the judges re-score; that's what the confidence label is for.

HOW THE SCORE WORKS

# Code checks what code can. Judges assess the rest.

### Measured by code

**Provenance:** is each value in the tool result it cites?

**Area:** is each site inside the boundary asked for?

+

### Three AI judges

8 criteria, from tool choice to honest uncertainty.

Opus 5.5 · DeepSeek V3.2 · Kimi K3

No judge grades its own family's agent.

→

### Median vote

Three votes, no ties: the median is always a real judge's vote.

→

### Score + confidence

**High** · Medium · **Low**: how far the judges agree.

No answer: **FAILED**.

Judges never see the code's results, so code and judges are two independent lines of evidence.

DOES THE EVALUATOR ITSELF WORK?

# We sabotaged runs on purpose. It noticed every time.

| PLANTED DEFECT | CRITERION IT SHOULD HIT | BEFORE → AFTER |
|---|---|---|
| Searches with no area | Tool arguments | 2 → 0 |
| A made-up site | Grounding | 2 → 0 |
| The answer removed | Completeness | 2 → 0 |

Live calibration, judges on a 0–2 scale. Its first pass caught a bug in our own evidence builder.

## Judges agree like experts do

**0.50**

AGREEMENT SCORE

Human experts in the same kind of study: **0.56**.

Krippendorff's α: 1 = perfect, 0 = chance.
95% CI 0.12–0.67 over 7 runs.

THE TRACES DON'T LIE

# It caught bugs in our own agent

BEFORE

## FAILED

Texas parks prompt: no answer, **$1.50** spent. The trace showed why, step by step.

AFTER THE FIXES

## 85.4 · High

Same prompt on AgentCore: 0 errors, 15 / 15 claims verified, 8 / 8 sites inside Texas, **$0.81**.

**"Northern California"** resolved to a 20-metre building in San Francisco, so every search came back empty.

**LGND sign-in** expired after exactly 300 seconds; every call after 290 s failed.

**Narrowed searches** (a 16 km circle around a park) were silently widened to all of Texas.

BUILT TO PLUG IN

# Every part is swappable by config

### Agent

Strands on Amazon Bedrock AgentCore; Claude Sonnet 5.5 or Kimi K3

### Tools (MCP)

AWS Open Data (RODA) and LGND Geo imagery search

### Trace

OpenTelemetry spans to an S3 archive; exporter swapped by config, proved in CI

### Evaluate

Measured checks + judge tribunal on Bedrock; Team Foundry console (FastAPI)

ADB'S RUBRIC → ONE YAML FILE

JUDGES → CONFIG

MODELS → CONFIG

TRACING BACKEND → CONFIG

INFRA → CDK

Replays work offline: any archived run re-scores from its trace, without calling the agent again.

WHAT IT PROVES, AND WHAT'S NEXT

# Honest about the limits

### It proves today

Each claim matches the tool output it cites, each site is in the requested area, and three independent judges agree how well the work was done.

### Not yet

Whether a site is **really** a data center. That needs an answer key, and ADB's own rubric.

### Next

An answer key (e.g. US solar sites), ADB's rubric as a file, expert ratings of the judges, plain-language verdicts, and checking every condition in the prompt.

Plain-language verdicts and prompt compliance are being built today, as separate slices.

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# Thank you. Questions?


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OPEN SOURCE · APACHE-2.0 github.com/onemorepereira/team46-breaking-barriers-hackathon